

PyPRIMAT

A Python package for precision Big Bang Nucleosynthesis

User guide and physics reference

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Abstract

PyPRIMAT is a Python re-implementation of the PRIMAT code for precise Big Bang Nucleosynthesis (BBN). It integrates the coupled equations governing (i) the cosmological background – photon and neutrino temperatures and the scale factor – and (ii) a nuclear reaction network, to predict the primordial abundances of ^1H , D , ^3He , ^4He , ^7Li and heavier nuclides. This document is meant to be *self-contained*: the main text shows how to *use* the code (Sec. 1) and gives the working equations for the three physical ingredients – the plasma thermodynamics (Sec. 2), the $n \leftrightarrow p$ weak interactions (Sec. 3) and the thermonuclear network (Sec. 4) – together with the run-time options that control them, followed by a section on the sensitivity of the abundances to the inputs (Sec. 5). All the lengthy formulas needed to actually reproduce the code – the full QED plasma corrections, every correction to the weak rates beyond the Born approximation, and the detailed-balance/NSE derivation – are collected in the appendices, so that the algorithms can be rewritten from this document alone. Equations follow the review of Pitrou, Coc, Uzan & Vangioni, *Physics Reports* **754** (2018) 1 [1] (cited as “Phys. Rep.”), and references therein, which the reader should consult for derivations. Figures are produced by the companion notebook `PyPRIMAT_doc_figures.ipynb`.

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1 Quick overview: using the code

PyPRIMAT can be driven in three equivalent ways, all of which build a parameter dictionary and call the top-level driver `pyprimat.PyPR`. Run from the repository root so that the data files under `rates/` resolve.

1.1 Command-line interface

The `pyprimat` console script exposes the most commonly varied options:

```
pyprimat --Omegabh2 0.02242 --network medium
```

which prints a short summary:

```
Neff = 3.04397730
YP (BBN) = 0.24691155
YP (CMB) = 0.24558556
D/H = 2.4381479e-05
He3/H = 1.0387101e-05
Li7/H = 5.489582e-10
--- running time: 1.22 seconds ---
```

The main flags are `--Omegabh2`, `--DeltaNeff`, `--network` (`small/medium/large` or any custom network file), `--amax`, `--numerical_precision`, `--json` (dump the full result dictionary), and `--verbose`.

1.2 Python API

For full control, build a parameter dictionary and call `PyPR` directly; every key of `config.DEFAULT_PARAMS` is accepted.

```
from pyprimat import PyPR

result = PyPR({"Omegabh2": 0.022425, "network": "medium"}).solve()
print(f"YP (BBN) = {result['YPBBN']:.8f}") # 0.24700173
```

```
print(f"D/H = {result['DoH']:.7e}") # 2.4356100e-05
print(f"Neff = {result['Neff']:.8f}") # 3.04397730
```

The `solve()` call returns a dictionary with `Neff`, the helium mass fractions `YPBBN/YPcmb`, the ratios `DoH`, `He3oH`, `He3oHe4`, `Li7oH`, and the neutrino observables `Omeganurel/OneOverOmeganunr`. The `PyPR` instance also exposes the background history and abundance trajectories as attributes used throughout the companion notebook: the temperature vectors `_Tg_vec`, `_Tnue_vec`, the weak-rate interpolants `_nT0p_frwr`, `_nT0p_bkwr` (functions of T_γ in *kelvin*), the compiled forward reaction rates `nucl.<reaction>_frwr` (function of T in kelvin, internally $T_9 = T/10^9$), the abundance interpolant `_Y_of_t`, and the final values `_Y_final`.

Output files. `output_time_evolution=True` writes `results/output_tables.tsv` with the full background and per-nuclide history; `output_final_result=True` writes the final mass fractions.

1.3 Graphical interface

After `pip install -e ".[gui]"`, a browser-based Streamlit app is available:

```
pyprimat-gui # console-script launcher
streamlit run pyprimat/gui/app.py # from a source checkout
```

It exposes the parameters as form fields and shows the final abundances and an interactive abundance-evolution plot.

1.4 Reading the numerical output

Several physical corrections shift N_{eff} at the 10^{-2} – 10^{-3} level, so abundances should be reported with enough significant figures: at least 8 decimals for N_{eff} and Y_P , 7 for D/H , and 6 for ${}^7\text{Li}/\text{H}$. A standard small-network run with $\Omega_b h^2 = 0.022425$ gives $Y_P \simeq 0.24700$, $\text{D}/\text{H} \simeq 2.4349 \times 10^{-5}$ and $N_{\text{eff}} \simeq 3.04398$.

2 Plasma thermodynamics

2.1 Background equations

The BBN background is a hot plasma of photons, electrons/positrons and three neutrino species in a spatially-flat Friedmann–Lemaître spacetime. Each species is described by a distribution function $f(t, p)$ obeying the Boltzmann equation in an expanding universe (Phys. Rep. Eq. 12),

$$\frac{\partial f}{\partial t} - H p \frac{\partial f}{\partial p} = C[f], \quad (1)$$

whose velocity moments give the conservation laws for number density and energy density (Phys. Rep. Eqs. 13, 16),

$$\dot{n} + 3Hn = \mathcal{J}, \quad \dot{\rho} + 3H(\rho + P) = \dot{q}, \quad (2)$$

with $\mathcal{J}$ and $\dot{q}$ the collisional source terms. The thermodynamic quantities are momentum integrals of f ; for a species of degeneracy g and mass m , writing $x \equiv m/T$, they reduce to the Fermi–Dirac (+) / Bose–Einstein (−) integrals $I_{\pm}^{(n,m)}$ defined in App. A (Phys. Rep. App. A):

$$\rho = \frac{g T^4}{2\pi^2} I_{\pm}^{(2,1)}(x), \quad P = \frac{g T^4}{6\pi^2} I_{\pm}^{(0,3)}(x), \quad s = \frac{\rho + P}{T}. \quad (3)$$

The photon gas ($g = 2$, $m = 0$) has $\rho_\gamma = (\pi^2/15) T_\gamma^4$. The expansion rate is fixed by the Friedmann equation (Phys. Rep. Eq. 9),

$$H^2 = \frac{8\pi G}{3} \rho_{\text{tot}}, \quad \rho_{\text{tot}} = \rho_\gamma + \rho_{e^\pm} + \sum_\nu \rho_\nu + \delta\rho_{\text{QED}} + \dots, \quad (4)$$

and the scale factor $a(T_\gamma)$ follows from conservation of the comoving electromagnetic-plasma entropy $\mathcal{S} a^3$ (Phys. Rep. Eqs. 21–24).

2.2 Solving the background: the two ODEs $a(T_\gamma)$ and $t(a)$

The code reduces the background to two ordinary differential equations, solved once at start-up (PyPR._setup_background_and_cosmo).

Scale factor from entropy conservation. Define the reduced plasma entropy $\mathcal{S}(T) \equiv \bar{s}_{\text{pl}}/\bar{s}_\gamma$ (with $\bar{s}_\gamma = 2\pi^2/45$, so $s_{\text{pl}} = \bar{s}_\gamma T^3 \mathcal{S}$); from Eq. (3), $\mathcal{S} = 1 + \frac{2}{\pi^2 \bar{s}_\gamma} [\frac{1}{3} I_+^{(0,3)}(x) + I_+^{(2,1)}(x)]$, running from $11/4$ at $T \gg m_e$ to 1 at $T \ll m_e$ (Phys. Rep. Eq. 28). Exact entropy conservation gives the algebraic relation $a(T) = a_0 T_0 / (T \mathcal{S}^{1/3})$ (Phys. Rep. Eq. 29). With incomplete neutrino decoupling the plasma loses entropy to the neutrinos at a dimensionless heating rate $\mathcal{N}(T)$ defined by $\dot{q}_{\text{pl}}/(HT) = -T^3 \mathcal{N}$ (Phys. Rep. Eq. 57), and the relation becomes the ODE actually integrated (Phys. Rep. Eq. 58),

$$\frac{d \ln a}{d \ln T} = - \frac{3 \bar{s}_{\text{pl}} + T d\bar{s}_{\text{pl}}/dT}{\mathcal{N} + 3 \bar{s}_{\text{pl}}}, \quad \bar{s}_{\text{pl}} \equiv \frac{s_{\text{pl}}(T)}{T^3}, \quad (5)$$

which reduces to $a \propto (T \mathcal{S}^{1/3})^{-1}$ when $\mathcal{N} = 0$ (the QED-corrected $\mathcal{S}^{\text{QED}}$ of App. B is used when QED is on). In the code $\bar{s}_{\text{pl}}$ and $d\bar{s}_{\text{pl}}/dT$ are returned together by `plasma.spl_and_dspl_dT`; Eq. (5) is `_dlnadlnT_NEVO`, integrated from T_{end} to T_{start} with `solve_ivp(method='LSODA')`, and inverted to $T(a)$. When `external_background=True`, $a(T)$ is instead read directly from the NEVO table's column $x \propto a$.

Time from the Friedmann equation. With $a(T)$ – hence $\rho_{\text{tot}}(T(a))$ – known, the Friedmann equation (4) is integrated as

$$\frac{dt}{d \ln a} = \frac{1}{H(a)}, \quad t_{\text{ini}} = \frac{1}{2H_{\text{ini}}} \quad (6)$$

(radiation-domination start), giving $t(a)$ and its inverse by interpolation (`_dtdlna`). All subsequent eras use the resulting $T(t)$, $a(t)$, $t(T)$. The plasma energy density entering H is $\rho_{\text{pl}} = \mathcal{E}(T) \bar{\rho}_\gamma T^4$ with $\mathcal{E} = 1 + \frac{30}{\pi^4} I_+^{(2,1)}(x)$ (plus the QED term $\delta g_\rho/2$), and the three neutrino flavours contribute $\rho_\nu = \bar{\rho}_\nu T_\nu^4$ each (Phys. Rep. Eqs. 30–31).

Baryon density and η_b . Baryons are cold, $\rho_b \propto a^{-3}$, so $n_b(a) = n_{b,0}/a^3$ with $n_{b,0} = \eta_b n_{\gamma,0}$ and the baryon-to-photon ratio $\eta_b = (\Omega_b h^2) \times C_\eta$, where the conversion factor $C_\eta = \rho_{100}^{\text{crit}}/(n_{\gamma,0} m_b)$ (`cfg.Omegabh2_to_eta0b`); numerically $\eta_b \simeq 6.09 \times 10^{-10} (\Omega_b h^2 / 0.02225)$ (Phys. Rep. Eqs. 33–35). This $n_b(a)$ sets the two-body fluxes $\Gamma = n_b \langle \sigma v \rangle$ of Sec. 4.

Optional energy-density components. Any extra contribution to ρ_{tot} in Eq. (4) enters through the `extra_rho` hook of PyPR (a list of functions $\rho(T_\gamma)$ [MeV⁴]). Two are built in: a constant extra radiation density from ΔN_{eff} decoupled species (`plasma.rho_nu_extra`, $\rho = \frac{7}{8} \frac{\pi^2}{15} T_\nu^4 \Delta N_{\text{eff}}$), and Early Dark Energy (`fEDE`, `zcEDE`, `wnEDE`; PyPR._setup_EDE) – a fluid peaking at redshift `zcEDE` with energy fraction `fEDE` and equation-of-state index `wnEDE`. Both modify only $H(T_\gamma)$, hence the time–temperature relation.

2.3 Neutrino decoupling and N_{eff}

Neutrinos decouple just before e^+e^- annihilation. In the idealized *instantaneous* limit, entropy conservation gives the asymptotic ratio $T_\nu/T_\gamma \rightarrow (4/11)^{1/3} \simeq 0.7138$, and the radiation density is parameterised by the effective number of neutrinos (Phys. Rep. Eq. 62),

$$\rho_{\text{rad}} = \rho_\gamma \left[1 + \frac{7}{8} \left(\frac{4}{11} \right)^{4/3} N_{\text{eff}} \right]. \quad (7)$$

In reality decoupling is *not* instantaneous: the highest-energy neutrinos are still coupled during annihilation and absorb a little of the released entropy, raising N_{eff} above 3. PyPRIMAT reads the precomputed non-instantaneous history $\{T_\nu^{(e,\mu,\tau)}(T_\gamma)\}$ from a NEVO table rather than recomputing the momentum-dependent Boltzmann hierarchy. Together with the QED plasma corrections (below) this gives the Standard-Model value $N_{\text{eff}} \simeq 3.044$ (Fig. 1).

2.4 QED plasma corrections

Electromagnetic interactions between the $e^\pm$ and the photon bath shift the effective electron and photon masses and hence the plasma equation of state (Phys. Rep. §II.E). To leading orders in the coupling the interaction-pressure correction is $\delta P = \delta P_d + \delta P_s$, with a dominant $O(\alpha)$ term and a subdominant two-loop exchange term; the explicit integrals (Phys. Rep. Eqs. 47–49) are given in App. B. The energy-density correction follows from the thermodynamic identity $\rho = -P + T dP/dT$, and the high-temperature limit is the constant

$$\delta g_\rho \simeq \delta g_P \simeq -\frac{25 \alpha_{\text{FS}}}{4\pi} \simeq -0.0145 \quad (8)$$

(Phys. Rep. Eq. 52). These corrections enter in four places (App. B): the plasma entropy $\mathcal{S}$ that sets $a(T_\gamma)$, the neutrino-temperature normalisation z_0 , the plasma energy density in Eq. (4), and – through the electron mass shift, which is bookkept as a finite-temperature weak-rate correction (App. C) – the weak rates.

2.5 In the code, with figures

Photon and neutrino thermodynamics are pure functions of temperature (`plasma.rho_g`, `plasma.rho_nu`); the $e^\pm$ quantities (`rho_e`, `p_e`) require the Fermi–Dirac integrals of Eq. (3) and are tabulated in an electron-thermo cache. The non-instantaneous $T_\nu(T_\gamma)$ history is read by `neutrino_history.py`; the QED corrections are loaded from `rates/plasma/QED_*.txt` or computed analytically by `qed_pressure.py`; the Friedmann rate is assembled in `PyPR._Hubble`. Figure 1 shows the per-flavour temperature ratios, Fig. 2 the effective relativistic degrees of freedom $g_{*\rho}, g_{*s}$ through annihilation, and Fig. 3 the QED pressure correction.

2.6 Associated options

Option	Default	Effect
<code>incomplete_decoupling</code>	True	Use the NEVO non-instantaneous $T_\nu(T_\gamma)$ table (else instantaneous).
<code>QED_corrections</code>	True	Include the QED plasma corrections (App. B).
<code>n_electron_table</code>	2000	Grid points for the $e^\pm$ thermodynamic tables.
<code>recompute_qed_corrections</code>	False	Recompute and overwrite <code>rates/plasma/QED_*.txt</code> .
<code>external_background</code>	False	Read $a(T_\gamma)$ directly from the NEVO table instead of solving the ODE.
<code>DeltaNeff</code>	0	Extra decoupled radiation added to ρ_{tot} .
<code>munuOverTnu</code>	0	Neutrino degeneracy $\xi_\nu = \mu_\nu/T_\nu$.

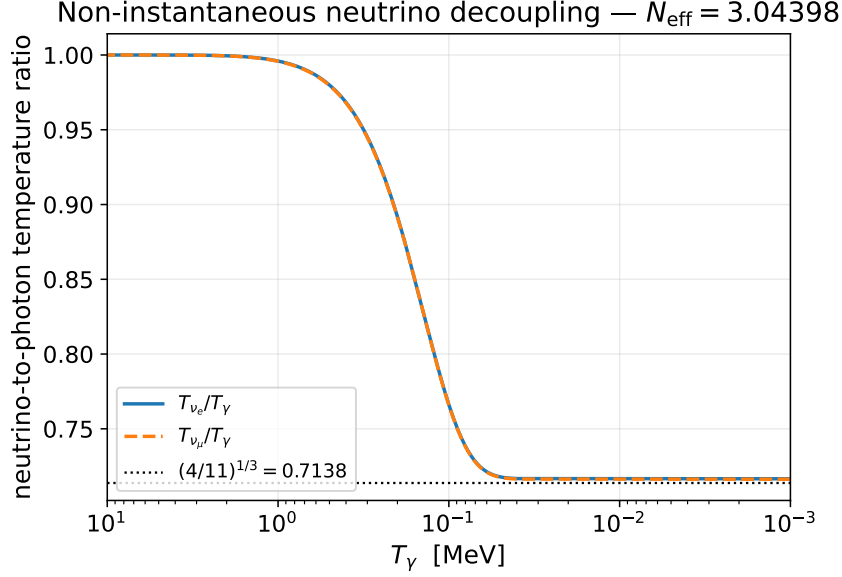


Figure 1: Per-flavour neutrino-to-photon temperature ratios from the NEVO non-instantaneous decoupling history. The dotted line is the instantaneous limit $(4/11)^{1/3}$; the residual offset is what raises N_{eff} above 3.

3 Weak interactions

3.1 The six processes and their rates

The neutron-to-proton ratio is controlled by six weak processes related by crossing symmetry (Phys. Rep. Eq. 67),

$$n + \nu_e \leftrightarrow p + e^-, \quad n \leftrightarrow p + e^- + \bar{\nu}_e, \quad n + e^+ \leftrightarrow p + \bar{\nu}_e, \quad (9)$$

grouped into a forward rate $\Gamma_{n \rightarrow p}$ (sum of the three) and a reverse $\Gamma_{p \rightarrow n}$. Each rate is a phase-space integral over the weak matrix element with the appropriate Fermi–Dirac occupations and Pauli-blocking factors (Phys. Rep. Eq. 70; App. C, Eq. (26)). The neutron-proton density evolution is

$$\dot{n}_n + 3Hn_n = -n_n\Gamma_{n \rightarrow p} + n_p\Gamma_{p \rightarrow n}. \quad (10)$$

3.2 Born approximation and calibration

In the infinite-nucleon-mass (Born) limit the matrix element collapses to a constant and the forward rate becomes the one-dimensional integral (Phys. Rep. Eqs. 77–83)

$$\bar{\Gamma}_{n \rightarrow p} = \mathcal{K} \int_0^\infty p^2 dp [\chi_+(E) + \chi_+(-E)], \quad \chi_\pm(E) \equiv (E_\nu^\mp)^2 g_\nu(E_\nu^\mp) g(-E), \quad (11)$$

with $E = \sqrt{p^2 + m_e^2}$, $E_\nu^\mp = E \mp \Delta$, $\Delta = m_n - m_p$, g and g_ν the electron and neutrino Fermi–Dirac distributions at T and T_ν , and the normalisation

$$\mathcal{K} = \frac{4G_W^2(1 + 3g_A^2)}{(2\pi)^3}, \quad G_W = G_F \cos \theta_C. \quad (12)$$

Because G_W and g_A carry $\sim 10^{-3}$ uncertainties, $\mathcal{K}$ is instead fixed by the measured free-neutron lifetime through $\mathcal{K} = 1/(\tau_n \lambda_0 m_e^5)$, where λ_0 is the dimensionless decay phase-space integral (Phys. Rep. Eqs. 89–91); its value rises from $\bar{\lambda}_0 \simeq 1.636$ (Born) to $\lambda_0^{\text{RC+FM}} \simeq 1.7547$ once corrections are included (App. C). When $T_\nu = T$ the rates obey the detailed-balance relation $\bar{\Gamma}_{p \rightarrow n} = e^{-\Delta/T} \bar{\Gamma}_{n \rightarrow p}$, which the code preserves order by order.

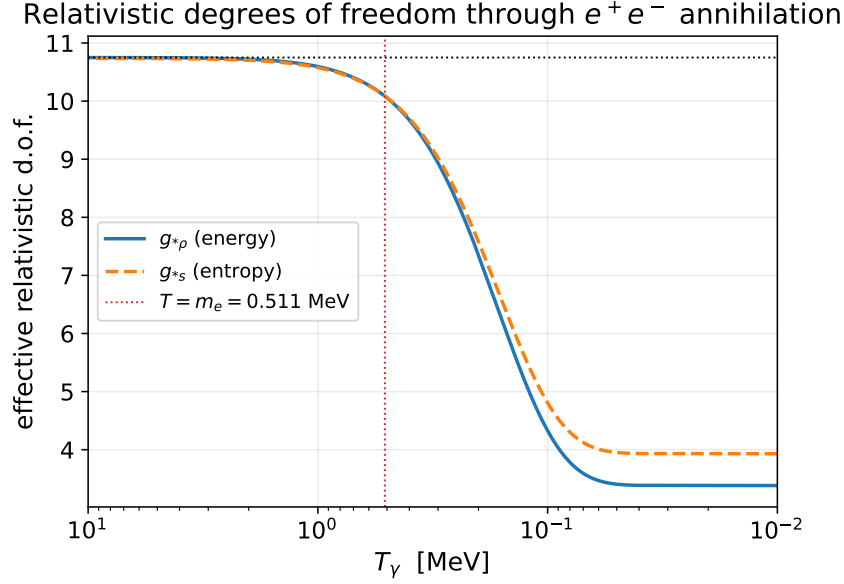


Figure 2: Effective relativistic degrees of freedom in energy ($g_{*\rho}$) and entropy (g_{*s}), dropping from 10.75 to $\simeq 3.36/3.93$ as e^+e^- annihilate around $T \sim m_e$, heating the photons relative to the neutrinos.

3.3 Corrections beyond Born

Reproducing Y_P to 10^{-3} requires several corrections, applied in sequence (full formulas in App. C):

1. *$T = 0$ radiative corrections*: a Coulomb (Fermi) factor $\mathcal{F}_{\pm}(E)$ multiplying the integrand when electron and proton are on the same side, times the resummed radiative factor $R(E, k_{\max}) = 1 + \frac{\alpha_{\text{FS}}}{2\pi} \mathcal{C}(E, k_{\max})$ built from Sirlin’s universal function (Phys. Rep. Eqs. 98–101; the Marciano–Sirlin resummation of [2]). Largest single effect, $\Delta Y_P \simeq +0.2\%$.
2. *Bremsstrahlung*: real-photon emission, added to keep detailed balance consistent with the finite- T terms (Phys. Rep. §III.E.2).
3. *Finite-temperature radiative corrections*: stimulated emission/absorption and the electron energy shift in the thermal bath (Brown & Sawyer [4]; Phys. Rep. §III.G), split into three pieces $\Gamma^{\gamma,T} + \Gamma^{\Delta E,T} + \Gamma^{ep+ee,T}$.
4. *Finite nucleon mass*: a Fokker–Planck expansion to first order in T/m_N giving $\chi_{\pm}^{\text{FM}}$ (Phys. Rep. §III.F; App. C).
5. *Weak magnetism*: at order T/m_N it is equivalent to a shift of the chiral couplings $c_{LL} \rightarrow c_{LL} + f_{\text{wm}} g_A$, $c_{RR} \rightarrow c_{RR} - f_{\text{wm}} g_A$ with $f_{\text{wm}} = (\mu_p - \mu_n)/2 \simeq 1.853$ (Phys. Rep. §III.F; Seckel [5]).
6. *Neutrino spectral distortions*: the equilibrium g_{ν} is replaced by the actual NEVO neutrino spectrum, which is slightly out of equilibrium.

Figure 4 compares the calibrated rates with H , and Fig. 5 decomposes the total correction to $\Gamma_{n \rightarrow p}$: the $T = 0$ radiative/finite-mass/weak-magnetism block dominates (~ -4 to -6%), while the finite- T and spectral pieces are sub-percent. Figure 6 shows the resulting neutron history.

3.4 In the code

`weak_rates.py` evaluates Eq. (11) and all corrections on the background grid. Results are cached in `rates/weak/` behind a configuration *fingerprint*, so a rerun with the same physics reloads

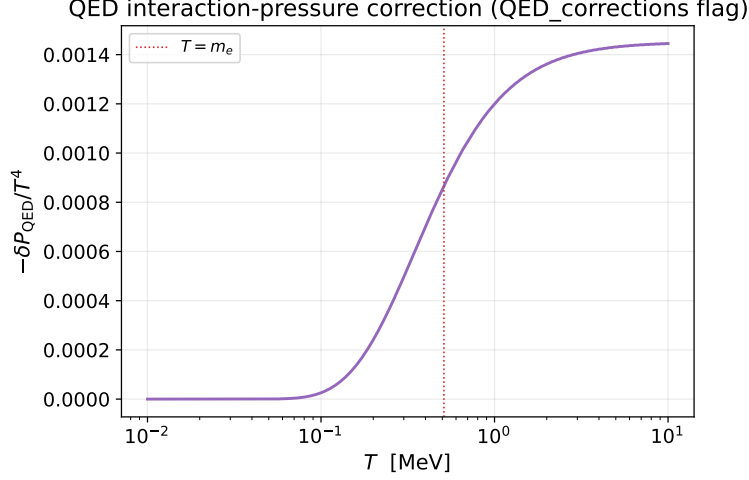


Figure 3: Leading QED interaction-pressure correction $-\delta P_{\text{QED}}/T^4$, peaking near $T \sim m_e$; its size is set by α_{FS}/π (toggled by `QED_corrections`).

them while a changed configuration recomputes. The normalisation $1/(\tau_n \lambda_0 m_e^5)$ is stored as `_NormWeakRates` and the temperature interpolants are `_nT0p_frwrdd/_nT0p_bkwrd` (argument in kelvin).

3.5 Associated options

Option	Default	Effect
<code>tau_n</code>	878.4	Neutron lifetime [s] used to calibrate $\mathcal{K}$.
<code>tau_n_flag</code>	True	Normalise via τ_n (else from G_F , V_{ud} , g_A).
<code>nT0p_Born_approximation</code>	False	Use only the Born rate (debugging/comparison).
<code>include_nT0p_thermal</code>	True	Apply the finite-temperature radiative correction.
<code>spectral_distortions</code>	True	Use the actual NEVO neutrino spectrum in the rates.
<code>sampling_nT0p</code>	200	Points in the $n \leftrightarrow p$ rate grid.
<code>weak_rate_cache</code>	True	Reuse the fingerprinted <code>rates/weak/</code> cache when valid.

4 Nuclear reactions

4.1 Thermal averaging of reaction rates

In the plasma the nuclei have a Maxwell–Boltzmann distribution of relative velocities (Phys. Rep. Eq. 118), and the thermonuclear rate of a two-body reaction is the thermal average of the cross-section (Phys. Rep. Eq. 119),

$$N_A \langle \sigma v \rangle = N_A \int_0^\infty \sigma(v) \phi_{\text{MB}}(v) v \, dv, \quad \phi_{\text{MB}}(v) v \, dv = \sqrt{\frac{8}{\pi m}} \frac{E e^{-E/k_B T}}{(k_B T)^{3/2}} \, dE, \quad (13)$$

in $\text{cm}^3 \text{s}^{-1} \text{mol}^{-1}$, with m the reduced mass. For charged-particle reactions σ is dominated by Coulomb-barrier tunnelling, so it is written with the astrophysical S -factor $\sigma(E) = S(E) e^{-2\pi\eta}/E$, where $\eta = Z_1 Z_2 e^2/(\hbar v)$ is the Sommerfeld parameter (Phys. Rep. Eqs. 120–122); $S(E)$ removes most of the energy dependence except for resonances. PyPRIMAT does not perform this average at run time: it reads the precomputed $N_A \langle \sigma v \rangle(T_9)$ tables (from the NACRE-II and related evaluations [8, 9]) and resamples them onto a common T_9 grid (log–log cubic interpolation). Five radiative-capture rates ($n(p, \gamma)d$, $d(p, \gamma)^3\text{He}$, $t(p, \gamma)^4\text{He}$, $t(\alpha, \gamma)^7\text{Li}$, $^3\text{He}(\alpha, \gamma)^7\text{Be}$) additionally receive a T_9 -dependent QED correction (`nuclear_qed_corrections`; Pitrou & Pospelov [10]),

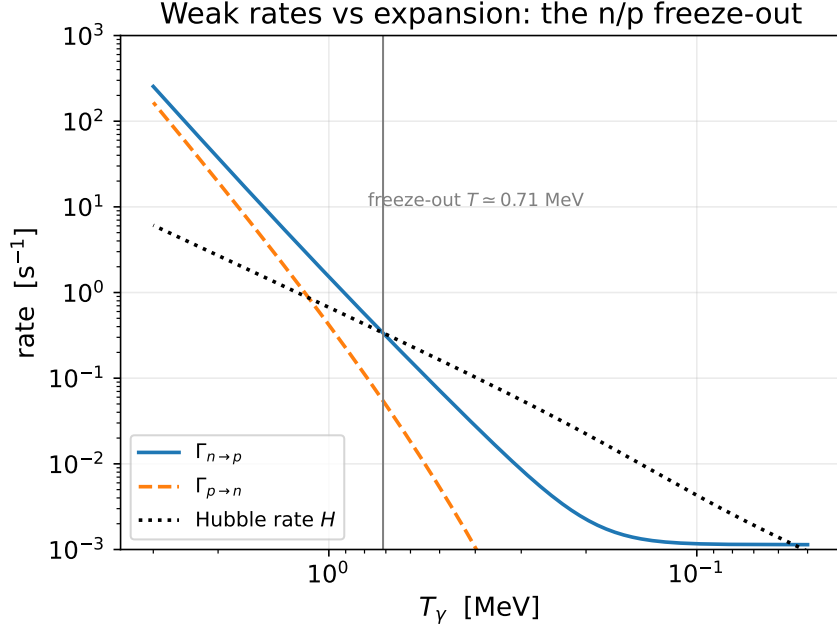


Figure 4: Calibrated weak rates and the Hubble rate H . Equilibrium holds where $\Gamma > H$; the high- T crossing is the n/p freeze-out. (Time flows left to right.)

applied as a multiplicative rescaling of the median rate at load time so that the corrected value becomes the new central rate.

4.2 The reaction network

With $Y_i \equiv n_i/n_b$ (so $\sum_i A_i Y_i = 1$), baryon conservation removes the dilution term and the abundances obey the Wagoner form (Phys. Rep. Eqs. 130–132)

$$\dot{Y}_i = \sum_{\text{reactions}} N_i \left[\Gamma_{j \rightarrow i} \frac{\prod_j Y_j^{N_j}}{\prod_j N_j!} - \Gamma_{i \rightarrow j} \frac{\prod_i Y_i^{N_i}}{\prod_i N_i!} \right], \quad (14)$$

where N_i is the stoichiometric coefficient of species i and the abundance rate is related to the number-density rate by $\Gamma_{i_1 \dots \rightarrow \dots} = n_b^{(\sum N_i)-1} \gamma$ with $\gamma = \langle \sigma v \rangle$. For the common two-body case $i + j \leftrightarrow k + l$ this is

$$\dot{Y}_i \supset Y_k Y_l \Gamma_{kl \rightarrow ij} - Y_i Y_j \Gamma_{ij \rightarrow kl}, \quad \Gamma_{ij \rightarrow kl} = n_b \langle \sigma v \rangle_{ij \rightarrow kl}. \quad (15)$$

Symmetry factors. The factorials $N_i!$ in Eq. (14) are the combinatorial symmetry factors that appear whenever a reaction has *identical* reactants or products. For example $d + d \rightarrow {}^3\text{He} + n$ has two identical deuterons in the entrance channel ($N_d = 2$): the number of distinct reacting pairs in a gas of n_d deuterons is $\frac{1}{2}n_d(n_d - 1) \simeq \frac{1}{2}n_d^2$, so the flux carries a factor $1/2! = 1/2$, and the stoichiometric prefactor $N_d = 2$ means each reaction removes two deuterons. The same $1/N!$ appears for identical products (e.g. ${}^7\text{Li} + p \rightarrow \alpha + \alpha$). Omitting these factors would double-count indistinguishable pairs.

4.3 Reverse rates from detailed balance

Only forward rates are tabulated; reverse rates are fixed by requiring that at nuclear statistical equilibrium (NSE) forward and reverse fluxes cancel. Using the equilibrium number densities

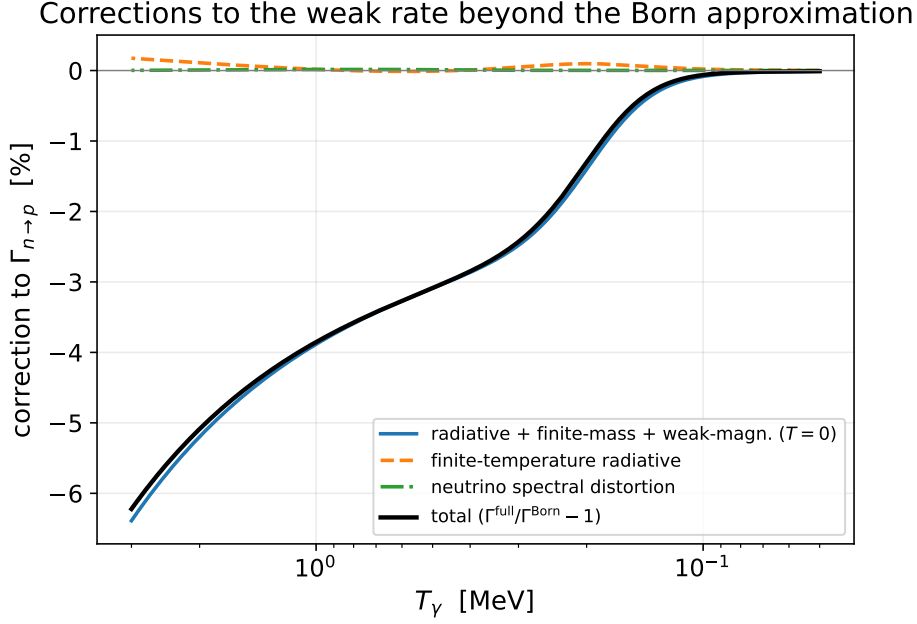


Figure 5: Decomposition of the correction to $\Gamma_{n \rightarrow p}$ relative to the Born rate, obtained by toggling the corresponding flags. The $T = 0$ block (radiative + finite-mass + weak-magnetism) dominates; finite-temperature and spectral-distortion contributions are at the per-mille level.

(App. D) the ratio depends only on the nuclide masses, spins and the reaction Q -value, and takes the three-parameter form (Phys. Rep. Eqs. 135–137)

$$\frac{\gamma_{kl \rightarrow ij}}{\gamma_{ij \rightarrow kl}} = \alpha \left(\frac{T}{10^9 \text{ K}} \right)^\beta \exp \left(\frac{\gamma \times 10^9 \text{ K}}{T} \right), \quad (16)$$

with (α, β, γ) computed once per reaction from nuclide data (`compute_detailed_balance_coefficients`). The derivation, including the symmetry factors and the binding-energy exponent, is given in App. D. Figure 7 shows several forward rates together with their reverse (photo-dissociation) rates: the reverse rate rises steeply with T_9 and crosses the forward rate around $T_9 \sim 1$ –2, i.e. where each capture is balanced by its inverse – this is the bottleneck that holds the nuclei at their NSE values until the universe cools enough.

4.4 The three-era ODE integration

The full network is far too stiff to integrate in one shot, so the solve is split at the fixed boundaries $T_{\text{start}}(10 \text{ MeV}) > T_{\text{weak}}(\sim 0.8 \text{ MeV}) > T_{\text{nucl}}(\sim 0.08 \text{ MeV}) > T_{\text{end}}(\sim 10^{-3} \text{ MeV})$, mapped to times through $t(T)$:

- **HT era** ($T_{\text{start}} \rightarrow T_{\text{weak}}$): only n, p , governed by the linear system $\dot{Y}_n = b Y_p - f Y_n$, $\dot{Y}_p = f Y_n - b Y_p$ with $f = \Gamma_{n \rightarrow p}$, $b = \Gamma_{p \rightarrow n}$, started from weak equilibrium $Y_n = b/(b + f)$ and integrated with the non-stiff LSODA solver.
- **MT era** ($T_{\text{weak}} \rightarrow T_{\text{nucl}}$): the fixed 18-reaction subset (even for medium/large – the full net is too stiff before the deuterium bottleneck opens), seeded with the HT $\{Y_n, Y_p\}$ and the NSE values Y_i^{NSE} (App. D) for all other species, integrated with the implicit BDF solver and the analytic Jacobian.
- **LT era** ($T_{\text{nucl}} \rightarrow T_{\text{end}}$): the full chosen network (12 / 62 / ~ 433 reactions), seeded from the MT result, BDF + Jacobian.

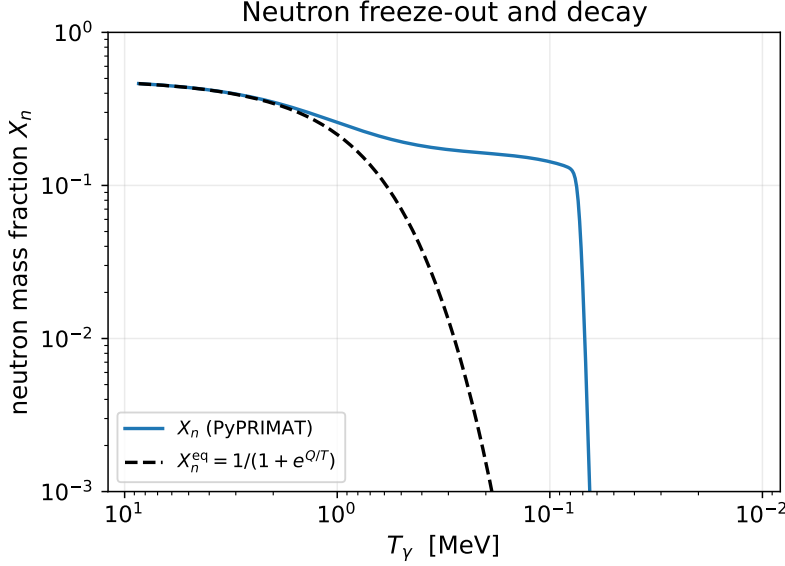


Figure 6: Neutron mass fraction X_n from PyPRIMAT (solid) versus the weak-equilibrium prediction $X_n^{\text{eq}} = 1/(1 + e^{Q/T})$ (dashed). X_n tracks equilibrium until freeze-out near 0.8 MeV, then declines slowly until deuterium formation burns the neutrons into ${}^4\text{He}$.

The normalised $n \leftrightarrow p$ rates feed all three eras as $f, b = \text{NormWR} \times \{\text{_nT0p_frwr}, \text{_nT0p_bkwr}\}$ with $\text{NormWR} = 1/(\tau_n F_n)$ in the `tau_n_flag` mode (F_n the decay phase-space factor). The reverse nuclear rates of Eq. (16) are clamped at low T_9 so that the exothermic factor e^{γ/T_9} cannot overflow (the “exothermic blow-up” guard in `nuclear.py`).

Solvers and the Jacobian. Two of `scipy.integrate.solve_ivp`’s integrators are used. **LSODA** (the Livermore solver with automatic stiffness detection) switches on the fly between a non-stiff Adams–Moulton predictor–corrector and a stiff backward formula; it is used for the smooth, mildly-stiff *background* ODEs [$a(T)$ of Eq. (5) and $t(a)$] and for the two-species HT system. The MT and LT *nuclear* networks are very stiff – a fast reaction and its detailed-balance reverse differ by many orders of magnitude, so the Jacobian has widely separated eigenvalues – and are integrated with the implicit **BDF** method (backward differentiation formulas), which is unconditionally stable for stiff systems where an explicit method would need prohibitively small steps. A BDF step solves a nonlinear system by Newton iteration and therefore needs the Jacobian $J_{ij} = \partial \dot{Y}_i / \partial Y_j$ of the right-hand side (14). Rather than let the solver estimate J by finite differences (costly and noisy for ~ 60 species), PyPRIMAT supplies it analytically: `compile_network` differentiates the polynomial mass-action terms symbolically and emits a fast `njit` kernel (`nucl.JacobianMT/JacobianLT`), passed to `solve_ivp` via its `jac` argument, which both accelerates and stabilises the integration.

4.5 Final observables

At T_{end} the code forms the nine entries of the result dictionary:

$$Y_P^{\text{BBN}} = 4Y_{{}^4\text{He}}, \quad \text{D/H} = \frac{Y_{\text{D}}}{Y_p}, \quad \frac{{}^3\text{He}}{\text{H}} = \frac{Y_{{}^3\text{H}} + Y_{{}^3\text{He}}}{Y_p}, \quad \frac{{}^7\text{Li}}{\text{H}} = \frac{Y_{{}^7\text{Li}} + Y_{{}^7\text{Be}}}{Y_p}, \quad (17)$$

where ${}^3\text{H} + {}^3\text{He}$ and ${}^7\text{Be} + {}^7\text{Li}$ include the post-BBN β -decays ${}^3\text{H} \rightarrow {}^3\text{He}$ and ${}^7\text{Be} \rightarrow {}^7\text{Li}$. N_{eff} is read off the neutrino-to-photon energy ratio at T_{end} via Eq. (7); Y_P^{CMB} is the helium mass fraction *by mass* (using the ${}^4\text{He}/{}^1\text{H}$ mass ratio) as relevant for CMB analyses; and `Omeganurel` / `OneOverOmeganunr` are the relativistic relic-neutrino $\Omega_\nu h^2$ and the inverse non-relativistic $\Omega_\nu h^2$ per summed eV, built from the T_ν/T_γ ratio at T_{end} scaled to today through T_0^{CMB} .

4.6 Networks, eras, and rate variation

The reaction list is named in `rates/nuclear/networks/{small,medium,large}.txt`; `load_network` parses it and `compile_network` turns the stoichiometry of Eq. (14) into fast right-hand-side and Jacobian kernels (the three eras of Sec. 4.4 share these kernels). Each reaction `<name>` carries a variation knob `p_<name>` sampling its rate at median $\times e^{p\sigma}$ for Monte-Carlo error propagation; the additive `NP_delta_<name>` (with `rescale_nuclear_rates=True`) multiplies it by $1 + \delta$:

```
PyPR({"network": "medium", "p_dpT0He3g": 1.0}).solve() # d(p,gamma)He3 at +1 sigma
```

Figure 8 shows the abundance history and Fig. 9 the final abundances versus the baryon density with the observational constraints.

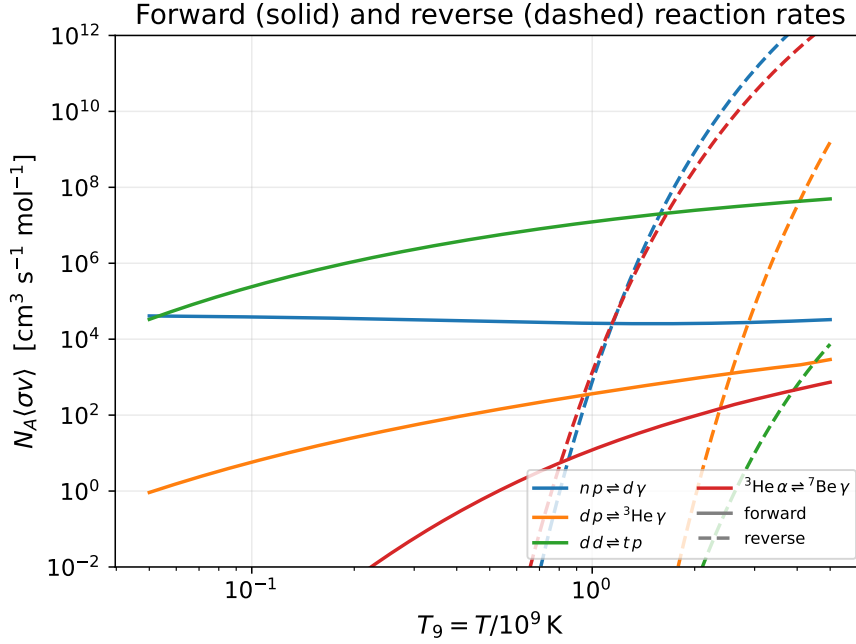


Figure 7: Forward (solid) and reverse (dashed, by detailed balance) rates $N_A\langle\sigma v\rangle(T_9)$ for four key reactions. The forward rates depend only weakly on T_9 ; the reverse photo-dissociation rates rise steeply and cross the forward rates near $T_9 \sim 1$ –2.

4.7 Rate uncertainties

Each evaluated rate comes with a 1σ uncertainty (a log-normal width σ per reaction). These propagate to the abundances; the dominant ones for Y_P are the neutron lifetime and the $n(p, \gamma)d$, $d(d, n)^3\text{He}$, $d(d, p)t$ reactions, with the rule-of-thumb (Phys. Rep. Eq. 145)

$$\Delta Y_P \approx 1.5 \times 10^{-3} \frac{\Delta\langle\sigma v\rangle}{\langle\sigma v\rangle}, \quad \Delta Y_P \approx 0.18 \frac{\Delta\tau_n}{\tau_n}, \quad (18)$$

while D/H is most sensitive to $d(p, \gamma)^3\text{He}$, $d(d, n)^3\text{He}$ and $d(d, p)t$. In PyPRIMAT the full budget is obtained by Monte-Carlo, drawing each `p_<rxn>` from $\mathcal{N}(0, 1)$; the helper `mc_uncertainty` parallelises this. Section 5 quantifies the individual responses.

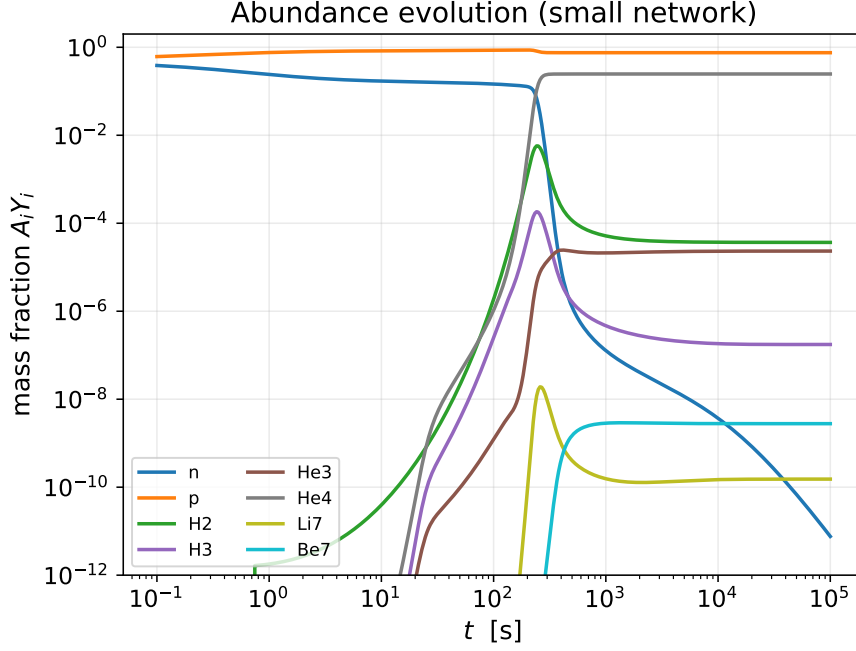


Figure 8: Mass fractions $A_i Y_i(t)$ for the small network. Deuterium builds up and is rapidly burned into ^4He once $T \sim 0.07$ MeV (the deuterium bottleneck), leaving the trace primordial D, ^3He , ^7Li and ^7Be .

4.8 Associated options

Option	Default	Effect
<code>network</code>	"small"	Reaction set: small (12), medium (62), large (~ 433).
<code>amax</code>	None	With large , drop nuclides of mass number $> A$.
<code>nuclear_qed_corrections</code>	True	QED corrections to radiative-capture rates [10].
<code>numerical_precision</code>	1e-7	Relative tolerance of the ODE solver.
<code>rate_grid_npts</code>	500	Points in the master T_9 grid for rate resampling.
<code>p_<reaction></code>	0	Log-normal variation of an individual rate (MCMC).
<code>output_time_evolution</code>	False	Write the full time series to <code>results/output_tables.tsv</code> .

5 Sensitivity of the abundances

A compact way to understand which inputs matter is the logarithmic sensitivity

$$S(O, p) \equiv \frac{\partial \ln O}{\partial \ln p} \approx \frac{\ln O(p(1 + \delta)) - \ln O(p(1 - \delta))}{2 \ln(1 + \delta)}, \quad (19)$$

the percentage change of observable O per percent change of parameter p , evaluated by a symmetric $\delta = 1\%$ finite difference. Figure 10 shows the matrix for the four light-element observables against the twelve key reaction rates and the physical parameters τ_n , G_N , $\Omega_b h^2$ and ΔN_{eff} . Reading it off: Y_P is almost insensitive to the nuclear rates but responds strongly to ΔN_{eff} (faster expansion $\rightarrow$ earlier freeze-out $\rightarrow$ more neutrons) and to τ_n ; D/H is driven by $\Omega_b h^2$, G_N and the deuterium-burning rates; $^7\text{Li}/\text{H}$ is anti-correlated with $\Omega_b h^2$ along the upper Schramm branch and sensitive to the $^3\text{He}(\alpha, \gamma)^7\text{Be}$ and $^7\text{Be}(n, p)^7\text{Li}$ rates. The figure reproduces `notebooks/Sensitivity.ipynb`.

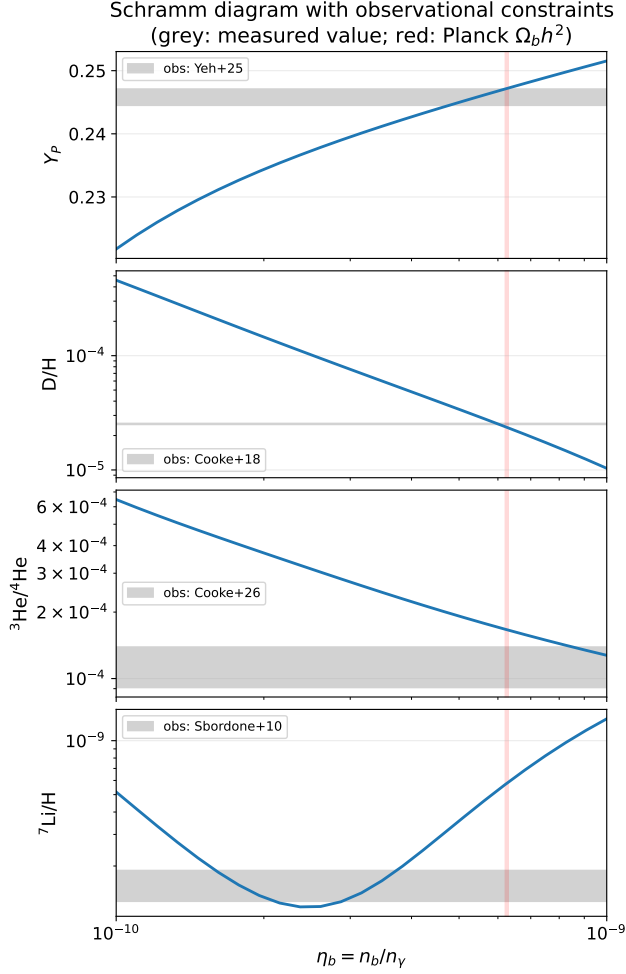


Figure 9: Schramm diagram with observational constraints (grey bands; values from `notebooks/StandardPlots.ipynb`). The red band is the Planck baryon density. Y_P and D/H agree with Planck, $^3\text{He}/^4\text{He}$ lies somewhat high, and $^7\text{Li}/H$ is over-predicted – the “lithium problem”.

A Thermodynamic integrals

The Fermi–Dirac (+) and Bose–Einstein (−) integrals used in Eq. (3) are (Phys. Rep. App. A)

$$I_{\pm}^{(n,m)}(x) \equiv \int_0^\infty \frac{p^n \mathcal{E}^{m-n}}{e^{\mathcal{E}} \pm 1} dp, \quad \mathcal{E} \equiv \sqrt{p^2 + x^2}, \quad x = m/T, \quad (20)$$

so that $\rho = \frac{gT^4}{2\pi^2} I_{\pm}^{(2,1)}$, $P = \frac{gT^4}{6\pi^2} I_{\pm}^{(0,3)}$ and $n = \frac{gT^3}{2\pi^2} I_{\pm}^{(2,0)}$. Useful limits are $I_+^{(2,1)}(0) = \frac{7\pi^4}{120}$, $I_+^{(0,1)}(0) = \frac{\pi^2}{12}$ and $I_-^{(0,1)} = \frac{\pi^2}{6}$. The electron chemical potential is non-zero only through the tiny charge-neutrality constraint $n_{e^-} - n_{e^+} = n_p$ and is neglected ($\xi_e \ll 1$) in the weak rates; it is kept only where it sets the absolute $e^\pm$ number density.

B QED plasma corrections

The thermal $e^\pm$ /photon interactions shift the masses by (Phys. Rep. Eqs. 40, 42; [6, 7])

$$\begin{aligned} \frac{\delta m_e^2(p, T)}{T^2} &= \frac{4\alpha_{\text{FS}}}{\pi} [I_-^{(0,1)} + I_+^{(0,1)}(x)] - \frac{2\bar{m}_e^2 \alpha_{\text{FS}}}{\pi \bar{p}} \int_0^\infty d\bar{q} \frac{\bar{q}}{\mathcal{E}_{\bar{q}}} \ln \left| \frac{\bar{p} + \bar{q}}{\bar{p} - \bar{q}} \right| \frac{1}{e^{\mathcal{E}_{\bar{q}}} + 1}, \\ \frac{\delta m_\gamma^2}{T^2} &= \frac{8\alpha_{\text{FS}}}{\pi} I_+^{(0,1)}(x), \end{aligned} \quad (21)$$

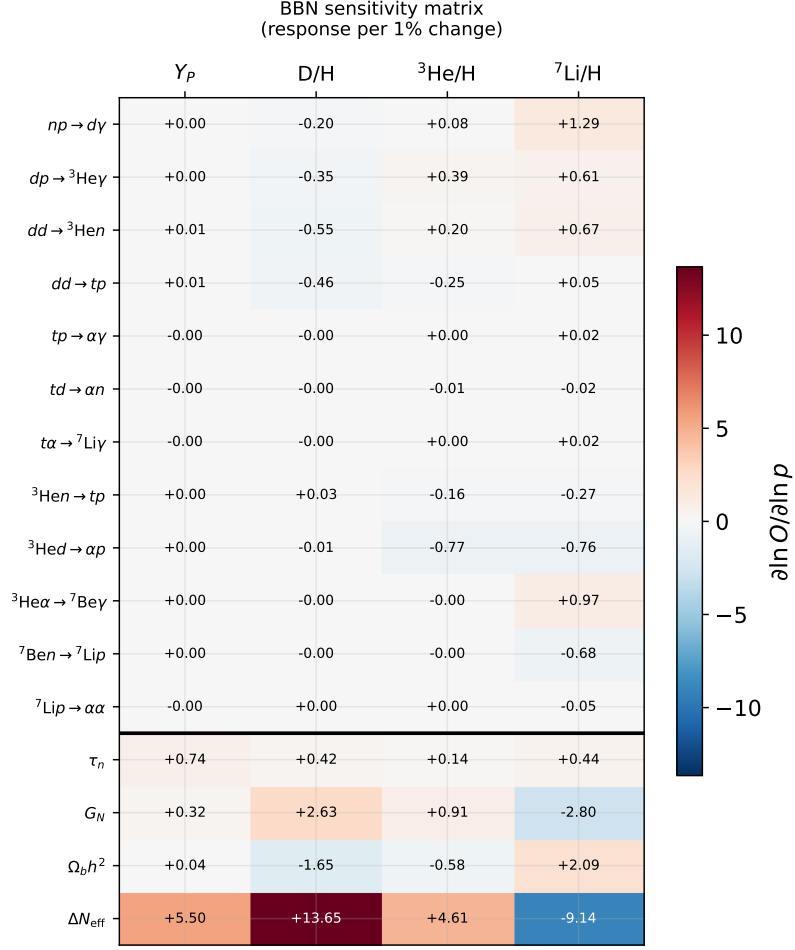


Figure 10: Logarithmic sensitivity $\partial \ln O / \partial \ln p$ of the four observables (columns) to the nuclear rates (top block) and physical parameters (bottom block), per 1% change. Red is positive, blue negative.

with $\bar{p} = p/T$, $x = \bar{m}_e = m_e/T$, $\mathcal{E}_{\bar{q}} = \sqrt{\bar{q}^2 + x^2}$. The induced pressure shift is a sum of three terms in increasing order of the electromagnetic coupling e (with $\alpha_{\text{FS}} = e^2/4\pi$). PyPRIMAT (`qed_pressure.py`) tabulates and *sums the first two*, stored as the two columns of `rates/plasma/QED_*.txt`:

$$\frac{\delta P_a}{T^4} = \frac{\alpha_{\text{FS}}}{\pi} \left[-\frac{2}{3} I_+^{(0,1)}(x) - \frac{2}{\pi^2} (I_+^{(0,1)}(x))^2 \right] \quad [O(e^2), \text{ dominant, negative}], \quad (22)$$

$$\frac{\delta P_{e^3}}{T^4} = \frac{4}{3} \sqrt{2\pi} \alpha_{\text{FS}}^{3/2} \left[\frac{I_+^{(0,1)}(x) + I_+^{(2,-1)}(x)}{\pi^2} \right]^{3/2} \quad [O(e^3), \text{ ring/plasmon}], \quad (23)$$

$$\frac{\delta P_b}{T^4} = \frac{\alpha_{\text{FS}}}{\pi^3} \bar{m}_e^2 \iint \frac{\bar{p}^2 d\bar{p}}{\mathcal{E}_{\bar{p}}} \frac{\bar{q}^2 d\bar{q}}{\mathcal{E}_{\bar{q}}} \ln \left| \frac{\bar{p} + \bar{q}}{\bar{p} - \bar{q}} \right| \frac{1}{(e^{\mathcal{E}_{\bar{p}}} + 1)(e^{\mathcal{E}_{\bar{q}}} + 1)} \quad [O(e^4), \text{ optional}], \quad (24)$$

with $I_+^{(2,-1)}(x) = \int_0^\infty \sqrt{p^2 + x^2} / (e^{\mathcal{E}} + 1) dp$ (I2m1). δP_a is the one-loop Frenkel–Galitskii–Migdal term, the $O(\alpha_{\text{FS}}^{3/2})$ ring term δP_{e^3} is the Blaizot–Zinn–Justin plasmon resummation (PRIMAT-Main.m), and the two-loop exchange δP_b – equal to the subdominant integral $\bar{\delta P}_s$ of Phys. Rep. Eq. 49 – is *off by default* (it would need a separate flag/file). The energy shift follows from $\rho = -P + T dP/dT$,

$$\bar{\rho} = 3\bar{\delta P} + \frac{\partial \bar{\delta P}}{\partial \ln T}, \quad \delta g_\rho \equiv \frac{30}{\pi^2} \bar{\delta \rho}, \quad \delta g_P \equiv \frac{90}{\pi^2} \bar{\delta P}, \quad (25)$$

with the high- T limit $\delta g_\rho \simeq \delta g_P \simeq -25\alpha_{\text{FS}}/(4\pi)$. These enter the code in four places (Phys. Rep. Eqs. 50–53): (i) the plasma entropy $\mathcal{S}^{\text{QED}} = \mathcal{S} + (\delta g_P + 3\delta g_\rho)/8$ used to obtain $a(T)$; (ii) the neutrino-temperature normalisation $(z_0^{\text{QED}})^3 = \frac{11}{4}(1 - \frac{4}{11}\frac{25\alpha_{\text{FS}}}{8\pi})$, giving $z_0^{\text{QED}} \simeq 1.39979$; (iii) the plasma energy density $\mathcal{E}^{\text{QED}} = \mathcal{E} + \delta g_\rho/2$ in Eq. (4); and (iv) through the electron mass shift – which is exactly the Brown–Sawyer relation $2E\Delta E = \delta m_e^2$ and is therefore counted as a finite-temperature weak-rate correction (App. C).

C Weak rates beyond the Born approximation

Matrix element

With the mostly-minus signature, each rate with nucleon a in the initial and b in the final state is (Phys. Rep. Eq. A40)

$$n_a \Gamma_{a \rightarrow b} = \int \frac{d^3 \mathbf{p}_a d^3 \mathbf{p}_e d^3 \mathbf{p}_\nu}{2^4 (2\pi)^8} \delta(E_a - E_b + s_e E_e + s_\nu E_\nu) \frac{|M|_{a \rightarrow b}^2}{E_n E_p E_e E_\nu} f_a f_\nu(s_\nu E_\nu) f_e(s_e E_e), \quad (26)$$

where $s_{e,\nu} = +1$ (-1) for an initial (final) lepton, and a final-state lepton enters as a Pauli-blocking factor via $f(-E) = 1 - f(E)$. The interaction is $\mathcal{H}_I = \frac{G_F}{\sqrt{2}} J_{e\nu}^\mu J_{pn,\mu}$ with a purely left-chiral leptonic current and the hadronic current (Phys. Rep. Eqs. A41–A44)

$$J_{pn}^\mu = \cos \theta_C \bar{p} \left(\gamma^\mu (1 - g_A \gamma^5) + i \frac{f_{\text{wm}}}{m_N} 2\Sigma^{\mu\nu} q_\nu \right) n, \quad f_{\text{wm}} = \frac{\mu_p - \mu_n}{2} \simeq 1.853. \quad (27)$$

Ignoring weak magnetism, the squared matrix element is $\frac{|M|^2}{2^7 G_F^2} = c_{LL} \mathcal{M}_{LL} + c_{RR} \mathcal{M}_{RR} + c_{LR} \mathcal{M}_{LR}$ with

$$c_{LL} = \frac{(1+g_A)^2}{4}, \quad c_{RR} = \frac{(1-g_A)^2}{4}, \quad c_{LR} = \frac{g_A^2 - 1}{4}, \quad (28)$$

and $\mathcal{M}_{LL} = (p_n \cdot p_\nu)(p_p \cdot p_e)$, $\mathcal{M}_{RR} = (p_n \cdot p_e)(p_p \cdot p_\nu)$, $\mathcal{M}_{LR} = m_p m_n (p_\nu \cdot p_e)$ (Phys. Rep. Eq. A46). All six processes of Eq. (9) share this matrix element (crossing + time reversal).

Born rate and calibration

In the infinite-mass limit the three $\mathcal{M}/(E_n E_p E_e E_\nu) \rightarrow 1$ (after an angular average), giving Eqs. (11)–(12). The decay phase-space factor is (Phys. Rep. Eq. 88)

$$\bar{\lambda}_0 = m_e^{-5} \int_0^{\sqrt{\Delta^2 - m_e^2}} p^2 E_\nu^2 dp = \sqrt{\bar{\Delta}^2 - 1} \frac{-8 - 9\bar{\Delta}^2 + 2\bar{\Delta}^4}{60} + \frac{\bar{\Delta}}{4} \text{arccosh } \bar{\Delta} \simeq 1.63609, \quad (29)$$

with $\bar{\Delta} = \Delta/m_e$; including radiative and finite-mass corrections raises it to $\lambda_0^{\text{RC+FM}} \simeq 1.75474$.

$T = 0$ radiative corrections

The corrected forward rate is (Phys. Rep. Eq. 99)

$$\Gamma_{n \rightarrow p}^{\text{RC0}} = \mathcal{K} \int_0^\infty p^2 dp [\mathcal{F}_+(E) R(E, E_\nu) \chi_+(E) + \mathcal{F}_+(-E) R(E, \Delta + E) \chi_+(-E)], \quad (30)$$

where the Fermi (Coulomb) factor $\mathcal{F}_\pm(x) = F(|x|)$ if $\pm x > 0$ and 1 otherwise switches on only when the electron and proton are on the same side. The radiative factor is $R(E, k_{\text{max}}) = 1 + \frac{\alpha_{\text{FS}}}{2\pi} \mathcal{C}(E, k_{\text{max}})$ with Sirlin’s function (Phys. Rep. Eqs. A78–A80; [3])

$$\mathcal{C}(E, k_{\text{max}}) = 4 \ln \frac{m_Z}{m_p} + \ln \frac{m_p}{m_A} + 2C + A_g + g(E, k_{\text{max}}), \quad (31)$$

$$g(E, k_{\text{max}}) = 3 \ln \frac{m_p}{m_e} - \frac{3}{4} + \frac{4}{\beta} L\left(\frac{2\beta}{1+\beta}\right) + 4[R(\beta) - 1] \left[\frac{k_{\text{max}}}{3E} - \frac{3}{2} + \ln \frac{2k_{\text{max}}}{m_e} \right] \\ + R(\beta) \left[2(1 + \beta^2) + \frac{k_{\text{max}}^2}{6E^2} - 4\beta R(\beta) \right], \quad (32)$$

with $\beta = p/E$, $R(\beta) = \frac{1}{2\beta} \ln \frac{1+\beta}{1-\beta}$, $L[x] = \int_0^x \frac{\ln(1-t)}{t} dt$ the Spence function, and the constants $C \simeq 0.891$, $A_g \simeq -0.34$, $m_A \simeq 1.2$ GeV. By default the code uses the Marciano–Sirlin resummed form (Phys. Rep. Eq. A81; [2])

$$R(E, k_{\max}) = \left[1 + \frac{\alpha_{\text{FS}}}{2\pi} (g(E, k_{\max}) - 3 \ln \frac{m_p}{2\Delta})\right] \left(L + \frac{\alpha_{\text{FS}}}{\pi} C + \frac{\alpha_{\text{FS}}}{2\pi} \delta\right) \left[S + \frac{\alpha_{\text{FS}}(m_p)}{2\pi} (\ln \frac{m_p}{m_A} + A_g) + \text{NLL}\right], \quad (33)$$

with $L \simeq 1.02094$, $S \simeq 1.02248$, $\alpha_{\text{FS}}(m_p) \simeq 1/134$. The reverse rate uses $\chi_+ \rightarrow \chi_-$, $\mathcal{F}_+ \rightarrow \mathcal{F}_-$, preserving detailed balance.

Bremsstrahlung and finite-temperature corrections

To keep detailed balance once the thermal bath is included, the neutron-beta-decay bremsstrahlung is re-added consistently (Phys. Rep. Eq. 100). The finite-temperature radiative correction has three parts (Phys. Rep. Eqs. 104–106; [4]), $\Gamma^T = \Gamma^{\gamma,T} + \Gamma^{\Delta E,T} + \Gamma^{ep+ee,T}$. The photon-exchange part is

$$\Gamma_{n \rightarrow p}^{\gamma,T} = \frac{\alpha_{\text{FS}} \mathcal{K}}{2\pi} \int_0^\infty \frac{dk}{k} \int_{m_e}^\infty dE \left\{ A(E, k) [g(-E) \tilde{\chi}_+^A(E, k) + g(E) \tilde{\chi}_+^A(-E, k)] - k B(E, k) [\dots] \right\}, \quad (34)$$

with $\tilde{\chi}_\pm^A(E, k) = \tilde{\chi}_\pm(E - k) + \tilde{\chi}_\pm(E + k) - 2\tilde{\chi}_\pm(E)$ and $\tilde{\chi}_\pm^B(E, k) = \pm[\tilde{\chi}_\pm(E - k) - \tilde{\chi}_\pm(E + k)]$; the electron-energy-shift part is

$$\Gamma_{n \rightarrow p}^{\Delta E,T} = -\frac{2\alpha_{\text{FS}} \mathcal{K}}{\pi} \int dp [\chi_+(E) + \chi_+(-E)] \left[\frac{\pi^2 T^2}{6} + \int_{m_e}^\infty dE' p' g(E') \left(1 - \frac{m_e^2}{E'^2 - E^2}\right) \right]. \quad (35)$$

These integrals are evaluated with the **vegas** Monte-Carlo library (or **scipy.dblquad**) and cached.

Finite nucleon mass and weak magnetism

The Born rate drops terms E_ν/m_N , E_e/m_N , Δ/m_N , all of order $\epsilon \equiv T/m_N \sim 10^{-3}$ near freeze-out. PyPRIMAT keeps them to first order in ϵ (neglecting $O(\epsilon^2) \sim 10^{-6}$) through a Fokker–Planck expansion of Eq. (26) in the energy transferred to the leptons, which adds (Phys. Rep. Eq. 94)

$$\delta\Gamma_{n \rightarrow p}^{\text{FM}} = \mathcal{K} \int_0^\infty p^2 dp [\chi_+^{\text{FM}}(E, g_A) + \chi_+^{\text{FM}}(-E, g_A)], \quad (36)$$

with $\Gamma_{p \rightarrow n}$ obtained by $\chi_+^{\text{FM}}(E, g_A) \rightarrow \chi_-^{\text{FM}}(E, -g_A)$. Because finite-mass and radiative corrections are both percent-level and do not add linearly, the multiplicative radiative factor $\mathcal{F}_\pm(E) R(E, k_{\max})$ is also applied to the finite-mass integrand ($\delta\Gamma^{\text{RC+FM}}$, Phys. Rep. Eq. 95). The kernel $\chi_\pm^{\text{FM}}(E, g_A)$ (Phys. Rep. Eq. A66) is built from the derivatives $g_\nu^{(n,p)}(E_\nu) \equiv \partial_{E_\nu}^p [E_\nu^n g_\nu(E_\nu)]$ and the reduced couplings $\tilde{c}_\pm = \frac{4}{1+3g_A^2} c_\pm$; its $T = 0$ limit (which enters the calibration λ_0) is

$$\chi_+^{\text{FM}}|_{T=0} = c_{\tilde{L}L} \frac{p^2 E_\nu^2}{m_N E} + c_{\tilde{R}R} \frac{E_\nu^3}{m_N} - (c_{\tilde{L}L} + c_{\tilde{R}R} + c_{\tilde{L}R}) \frac{2E_\nu^3 + E_\nu p^2}{m_N} + c_{\tilde{L}R} \frac{p^2 E_\nu^2}{m_N E}. \quad (37)$$

Weak magnetism, at this order, is equivalent to the shift

$$c_{LL} \rightarrow \frac{(1 + g_A)^2}{4} + f_{\text{wm}} g_A, \quad c_{RR} \rightarrow \frac{(1 - g_A)^2}{4} - f_{\text{wm}} g_A, \quad c_{LR} \text{ unchanged}, \quad (38)$$

so it is implemented simply by modifying the couplings entering $\chi_\pm^{\text{FM}}$ (Phys. Rep. §III.F; [5]). Combined with radiative corrections it shifts λ_0 by $\delta\lambda_0^{\text{FM}} \simeq -3.63 \times 10^{-3}$.

Spectral distortions

Finally, the equilibrium g_ν in $\chi_\pm$ is replaced by the actual neutrino occupation $f_\nu = g_\nu + \delta f_\nu$, which deviates from a Fermi–Dirac because of incomplete decoupling (`spectral_distortions=True`). PyPRIMAT supports two forms of δf_ν (`neutrino_history.py`):

- *NEVO-table* (default, `analytic_distortions=False`): the distortion is read from the full NEVO spectrum file as a function of comoving momentum $y = E_\nu/T_\nu$ and T_γ .
- *Analytic μ/y* (`analytic_distortions=True`; requires instantaneous decoupling), two continuous knobs `delta_xi_nu` and `y_SZ`:

$$\delta f_\nu^\mu(y) = \frac{1}{e^{y-(\xi+\delta\xi)} + 1} - \frac{1}{e^{y-\xi} + 1} \quad (\mu\text{-type, shift of } \xi = \mu_\nu/T_\nu), \quad (39)$$

$$\delta f_\nu^y(y) = f_{\text{FD}}(1 - f_{\text{FD}}) [4 - y(1 - 2f_{\text{FD}})] \quad (y\text{-type / SZ}), \quad (40)$$

where $f_{\text{FD}} = 1/(e^y + 1)$; the y -type shape is $y^{-2} d_y(y^4 d_y f_{\text{FD}})$, the Sunyaev–Zel’dovich form. The same distortion adds an energy density ρ_ν^{SD} to the Friedmann equation (4).

D Detailed balance and nuclear statistical equilibrium

At thermal equilibrium a non-relativistic species has number density (Phys. Rep. Eq. A33)

$$n_i = g_i \left(\frac{m_i T}{2\pi} \right)^{3/2} e^{\xi_i - x_i}, \quad g_i = 2s_i + 1, \quad \xi_i = \mu_i/T, \quad x_i = m_i/T. \quad (41)$$

At NSE a nuclide is in chemical equilibrium with its nucleons, $\mu_i = Z_i \mu_p + (A_i - Z_i) \mu_n$, which gives (Phys. Rep. Eq. A35)

$$n_i^{\text{NSE}} = \frac{g_i m_i^{3/2}}{2^{A_i}} \left(\frac{n_p}{m_p^{3/2}} \right)^{Z_i} \left(\frac{n_n}{m_n^{3/2}} \right)^{A_i - Z_i} \left(\frac{2\pi}{T} \right)^{3(A_i - 1)/2} e^{B_i/T}, \quad B_i = Z_i m_p + (A_i - Z_i) m_n - m_i, \quad (42)$$

with B_i the binding energy. Requiring forward and reverse fluxes to cancel at NSE, the general detailed-balance relation is (Phys. Rep. Eq. 133)

$$\frac{\gamma_{i_1 \dots i_p \rightarrow j_1 \dots j_q}}{\gamma_{j_1 \dots j_q \rightarrow i_1 \dots i_p}} = \frac{N_{i_1}! \dots N_{i_p}!}{N_{j_1}! \dots N_{j_q}!} \left[\frac{n_{j_1}^{N_{j_1}} \dots}{n_{i_1}^{N_{i_1}} \dots} \right]^{\text{NSE}}, \quad (43)$$

where the factorials are exactly the symmetry factors of Eq. (14). Substituting n_i^{NSE} and using that nuclear reactions conserve Z and A (so the n_p, n_n powers cancel) leaves a function of masses, spins and T only (Phys. Rep. Eq. 134),

$$\frac{\gamma_{j \rightarrow i}}{\gamma_{i \rightarrow j}} = \frac{\prod_i \frac{1}{N_i!} [g_i (m_i T / 2\pi)^{3/2}]^{N_i}}{\prod_j \frac{1}{N_j!} [g_j (m_j T / 2\pi)^{3/2}]^{N_j}} \exp\left(\frac{\sum_j m_j - \sum_i m_i}{T} \right), \quad (44)$$

which collapses to the three-constant form of Eq. (16): α gathers the spin/mass/ 2π prefactors, $\beta = \frac{3}{2}(\sum N_i - \sum N_j)$ the power of T , and $\gamma \propto$ the reaction Q -value $\sum_j m_j - \sum_i m_i$. The masses and spins are taken from the AME/NUBASE tables [11]. PyPRIMAT computes (α, β, γ) once per reaction in `compute_detailed_balance_coefficients` and stores them in `rates/nuclear/data/detailed_balance.csv`.

E Reproducing the figures

Every figure in this document is generated by the notebook `doc/PyPRIMAT_doc_figures.ipynb`, which writes PDFs into `doc/figures/`. To refresh them after changing the code, run the notebook top to bottom, or non-interactively:

```
jupyter nbconvert --to notebook --execute --inplace PyPRIMAT_doc_figures.ipynb
```

and recompile. The notebook uses only the public PyPRIMAT API and doubles as worked examples for extracting the background history, weak rates, reaction rates and abundance trajectories from a run.

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